AQ400 Method

EPA-118-C Rev. 2

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o-Phosphate–P in Drinking, Saline and Surface Waters, and Domestic and Industrial Wastes

1 SCOPE

This method covers the determination of o-Phosphate–P in surface and saline waters, drinking waters and domestic and industrial waste waters. Samples which have been digested according to EPA methods 365.1, 365.2 or 365.3 and are neutralized may also be analyzed using this method.

This method is approved for the Clean Water Act for use in wastewater compliance monitoring, under National Pollutant Discharge Elimination System (NPDES). This method is equivalent to EPA Method 365.1, Rev. 2.0 (1993), Standard Methods 4500-P-F (1999) and USGS Method I-4601-85 and is approved for the Safe Drinking Water Act for use in drinking water compliance monitoring, under National Primary Drinking Water Regulations (NPDWR).

2 RANGE OF APPLICATION

Range	0.005 – 1.0 mg P/L

3 METHOD DETECTION LIMIT

By EPA Procedure	MDL = 0.0015 mg P/L

4 METHOD PRINCIPLE

Reaction with acidic molybdate in the presence of antimony forms an antimony phospho-molybdate complex, which then reduced by ascorbic acid to phosphomolybdenum blue and is measured photometrically at 880 nm.

5 REFERENCES

Methods for the Determination of Inorganic Substances in Environmental Samples, USEPA 600/R 93/100, August 1993: Method 365.1, Rev. 2.0.

Standard Methods for the Examination of Water and Wastewater, APHA/AWWA/WEF, method 4500-P-F (1999 forward).

6 INTERFERENCES

Interferences can come from a variety of analytes including barium, lead, iron, silver, silica, nitrite, sulfide, arsenate, hexavalent chromium, and arsenic; however the effects of these are usually negligible in natural waters. See Section 11.1, Chemistry Troubleshooting Tips for specifics.

Salt errors for samples ranging from 5-20% salt content were found to be less than 1%.

Arsenate is determined similarly to phosphorus and should be considered when present in concentrations higher than phosphorus. However, at concentrations found in sea water, it does not interfere.

Filtration of samples for total or total hydrolysable phosphorus should only occur after digestion.

Bias from sample color or turbidity is correctable, using the sample blanking feature of AQ software.

7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the material safety data sheets (SDS) for details on safe handling of chemicals.

Chemical	Formula	C.A.S. No.	Safety Classification
Ammonium molybdate tetrahydrate	$(NH_4)_6Mo_7O_{24} \bullet 4H_2O$	12054-85-2	--
Ascorbic acid	$C_6H_8O_6$	50-81-7	--
EDTA disodium salt dihydrate*	$C_{10}H_{14}N_2Na_2O_8 \bullet 2H_2O$	6381-92-6	GHS07, GHS09
Potassium antimonyl tartrate trihydrate	$C_8H_4K_2O_{12}Sb_2 \bullet 3H_2O$	28300-74-5	GHS07, GHS09
Potassium phosphate monobasic	KH_2PO_4	7778-77-0	--
Sodium hydroxide*	NaOH	1310-73-2	GHS05
Sulfuric acid, concentrated	H_2SO_4	7664-93-9	GHS05

GHS01: Danger – Explosive	GHS02: Danger – Flammable	GHS03: Danger – Oxidizing
GHS04: Warning – Compressed Gas	GHS05: Warning/Danger – Corrosive	GHS06: Danger – Toxic
GHS07: Warning – Irritant	GHS08: Danger – Health Hazard	GHS09: Warning/Danger – Aquatic Toxicity

Note: Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

* Used only for Special Cleaning Solution

8 REAGENT SOLUTIONS

Use reagent-grade chemicals, certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination, therefore Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

8.1 SULFURIC ACID, 5 normal

Sulfuric acid, concentrated	70 mL
DI Water	dilute to 500 mL

Slowly add 70 mL concentrated sulfuric acid to about 400 mL of deionized water. **CAUTION**, the container will become very warm. Cool to room temperature and dilute to 500 mL with deionized water.

Store at room temperature. This solution is available commercially. Stable for 6 months.

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8.2 POTASSIUM ANTIMONYL TARTRATE, 3 g/L

Potassium antimonyl tartrate trihydrate	1.5 g
DI water	dilute to 500 mL

Dissolve 1.5 g potassium antimonyl tartrate trihydrate in about 400 mL deionized water. Dilute to 500 mL and mix well.

Store at 4°C in a plastic bottle. Stable for 1 month.

8.3 AMMONIUM MOLYBDATE, 4% (w/v)

Ammonium molybdate tetrahydrate	4.0 g
DI water	dilute to 100 mL

Dissolve 4 g ammonium molybdate tetrahydrate, in approximately 80 mL deionized water. Stir at least 1 hour. Dilute to 100 mL.

Store at 4°C in a plastic bottle. Stable for 1 month. Discard this reagent if it becomes discolored or turbid.

8.4 WORKING PHOSPHATE COLOR REAGENT

Sulfuric acid, 5 normal (above)	65 mL
Potassium antimonyl tartrate, 3 g/L (above)	7.5 mL
Ammonium molybdate, 4% w/v (above)	22 mL
DI water	dilute to 100 mL

To a 100 mL volumetric flask, add 65 mL sulfuric acid, 5 normal followed by 7.5 mL potassium antimonyl tartrate, 3 g/L and swirl to mix. Then add 22 mL ammonium molybdate, 4%. Swirl the contents, fill the flask to the mark with deionized water and mix well.

Store at 4°C in a plastic container. Stable for 3 weeks. Discard if the reagent turns blue or becomes turbid.

8.5 WORKING ASCORBIC ACID, 15 g/L (with o-phosphate spike)

Ascorbic acid, fine granular	1.5 g
ortho-Phosphate spike	0.12 mg P/L
DI water	dilute to 100 mL

Dissolve 1.5 g of ascorbic acid, fine granular, in about 80 mL deionized water. Spike this reagent with o-phosphate to a level about 0.12 mg P/L. For example, add 0.24 mL of 50 mg P/L o-phosphate standard to 100 mL to produce a spike level of 0.12 mg P/L. Dilute to 100 mL and mix well.

Store at 4°C in a plastic bottle. Stable for 2 days, but for best results prepare fresh daily.

8.6 SPECIAL CLEANING SOLUTION

Sodium hydroxide	4.0g
EDTA disodium salt dihydrate	1.0 g
DI Water	100 mL

Dissolve 4.0 g of sodium hydroxide in approximately 80 mL of water. Add 1.0 g of EDTA disodium salt dihydrate. Dilute to 100 mL and mix well.

Store at room temperature. Stable for 2 months.

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9 STANDARD SOLUTIONS

9.1 PHOSPHATE STOCK STANDARD SOLUTION (1000 mg P/L)

Potassium phosphate monobasic	4.394 g
DI water	dilute to 1 L

Dissolve 4.394 g potassium dihydrogen orthophosphate, anhydrous (dried at 105°C) in about 800 mL deionized water in a 1 Liter volumetric flask. Up to 10 mL sulfuric acid, 5 normal, may be added for preservation. Dilute to the mark with deionized water.

Store at 4°C. Stable for 1 month.

9.2 CALIBRATION DILUENT (for acid-preserved samples)

Sulfuric acid, 5 normal (above)	see table
DI water	dilute to 2 L

Use this solution as the diluent for automated and manual dilutions and also as the blank and diluent for Auto-calibrations. Note that use of sulfuric acid for preservation of field samples is not allowed for compliance monitoring analysis of o-phosphate in drinking waters. See section 10 for details.

Choose the level of sulfuric acid closest to the level of acid added to field samples for preservation. Transfer the volume of 5 normal sulfuric acid shown in the table (±0.25 mL) to a 2 Liter bottle. Dilute to volume with deionized water.

Store up to 6 months.

Sulfuric acid, preservation level	None	1.0 mL/L	1.5 mL/L	2.0 mL/L
5 normal sulfuric acid, volume added per 2 L	--	14 mL	21 mL	29 mL

9.3 PHOSPHATE INTERMEDIATE STANDARD SOLUTION (50 mg P/L)

PHOSPHATE Stock Standard (1000 mg P/L, above)	10 mL
Calibration diluent (above)	dilute to 200 mL

Pipet 10 mL PHOSPHATE–P Stock Standard (1000 mg P/L) to a 200 mL volumetric flask. Fill to the mark with Calibration Diluent and invert to mix.

Store at 4°C. Stable for 2 weeks.

9.4 AUTO CALIBRATION STANDARD – Phosphate-P 1.0 mg P/L Solution

PHOSPHATE Intermediate Standard (50 mg P/L, above)	5 mL
Calibration diluent (above)	dilute to 250 mL

Transfer 5 mL PHOSPHATE–P Intermediate Standard (50 mg P/L) to a 250 mL volumetric flask. Fill to the mark with Calibration diluent and invert to mix.

Store at 4°C. Prepare fresh weekly.

9.5 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

Concentration, mg P/L	0	0.01	0.04	0.1	0.4	0.8	1.0
Volume (mL) Auto Calibration Standard Solution diluted as indicated with Calibration diluent	--	2	10	10	20	40	50
Final Volume, mL	100	200	250	100	50	50	50

Prepare weekly.

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10 SAMPLE PRESERVATION AND STORAGE

Samples may be collected into plastic containers.

Samples to be reported for ortho-phosphate for EPA's drinking water compliance monitoring programs should NOT be filtered, digested, or undergo hydrolysis. The samples should be cooled to ≤ 6°C and must be analyzed within 48 hours of collection. Do not use acid for preservation.

Samples to be reported for dissolved ortho-phosphate for EPA's wastewater compliance monitoring programs must be filtered immediately, then cooled to ≤ 6°C and must be analyzed within 48 hours of collection. Do not use acid for preservation.

Samples to be reported for total-phosphorus for EPA's wastewater compliance monitoring programs must be preserved with sulfuric acid to a pH < 2, cooled to ≤ 6°C and must be analyzed within 28 days. Digestion is required as described in EPA method 365.1, 365.2 or 365.3. Do not filter prior to digestion. After digestion, neutralize with sodium hydroxide as described in EPA method 365.1 or 365.2. If pH neutralization is not desired, analyze the finished digests using SEAL method EPA-119-C instead. Call SEAL Analytical Technical Support for assistance.

11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method are only suggestions.

Any reagents having visible particulates should be filtered.

Pour calibration standards, quality control solutions and samples into sample cups and reagent vessels, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent vessels and a waste collection bottle in place.

After o-phosphate analyses are finished for the day, clean the cuvette to remove deposits. Fill a reagent wedge with SEAL Cuvette Cleaning Solution (p/n 5676) and place it in Position 1 of the reagent rack. Within SEAL discrete analyzer software, go to Maintenance–Cuvette Functions and select Extra Wash.

11.1 CHEMISTRY TROUBLESHOOTING TIP

Glassware should be washed with 1:1 HCl, rinsed with phosphate color reagent, and rinsed thoroughly with deionized water. If glassware is reserved solely for the determination of phosphorus, then treatment with 1:1 HCl and phosphate color reagent is only required occasionally. Commercial detergent should never be used.

As much as 50 mg Fe^{3+}/L, 10 mg Cu^{2+}/L, and 10 mg SiO_2/L can be tolerated. High silica concentrations cause positive interference and high iron concentrations can cause precipitation of, which causes a subsequent loss of phosphorus. For silica concentrations of 20, 50, and 100 mg/L, phosphorous results will be elevated by 0.0005, 0.015, and 0.025 mg/L, respectively.

Eliminate interferences from nitrite and sulfide by adding an excess of bromine water or a saturated potassium permanganate solution.

Hexavalent chromium and nitrite interfere to give results about 3% low at concentrations of 1 mg/L and 10 to 15% low at 10 mg/L. Sulfide (Na_2S) and silicate do not interfere at concentrations of 1.0 and 10 mg/L, respectively.

The molybdate to acid ratio is very important for the phosphate chemistry. If this ratio is off, then the curve will be quadratic. If curve is quadratic, check age of the acid and molybdate solutions. Remake color reagent and stocks.

If running a lot of phosphate on the AQ400, it is recommended to run the Special Cleaning Solution (section 8.6) to clean the cuvette. To do this, fill a reagent wedge with Special Cleaning Solution and place it in Position 1 of the reagent rack. Within SEAL AQ software, go to Maintenance—Cuvette Functions and select Extra Wash.

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12 TEST PARAMETERS

PARAMETER	AQ SETTING
Test name	o-PHOS 1
Units	mg P/L
Decimals	3
Test type	End Point
Sample volume (µL)	400
Water volume (µL)	0
Number of mixes	2
Cuvette primes	4
Cuvette washes	2
Baseline on Wash	Ticked
Reaction time (seconds)	480
Wavelength (nm)	880
Polynomial order	1
Number of reagents	2
1. Working phosphate color reagent (µL)	140
2. Working ascorbic acid (µL)	45

DEFINITIONS:

Test name:	Name which appears on the final report
Units:	Concentration units appearing against the reported result
Decimals:	Number of decimal places in the reported result
Test type:	The type of test the software will perform
Sample volume:	Volume (in microliters) of the sample used in the test
Water volume:	Volume (in microliters) of water from the water reservoir used in the test
Number of mixes:	Number of mixes of the sample + reagent in the reaction well
Cuvette primes:	Number of times the cuvette is flushed with reagent (no sample) prior to sampling
Cuvette washes:	Number of washes of the cuvette between aspirations of finished reaction liquid
Baseline on Wash:	Water baseline measured between aspirations to provide drift correction
Reaction time:	Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture
Wavelength:	Interference filter wavelength to be used in the test
Polynomial order:	Polynomial order to be used when fitting the standard points to the standard graph (1 = linear)
Number of reagents:	Number of reagents defined for the test
Reagent volume:	Volume (in microliters) of the reagent used in the test

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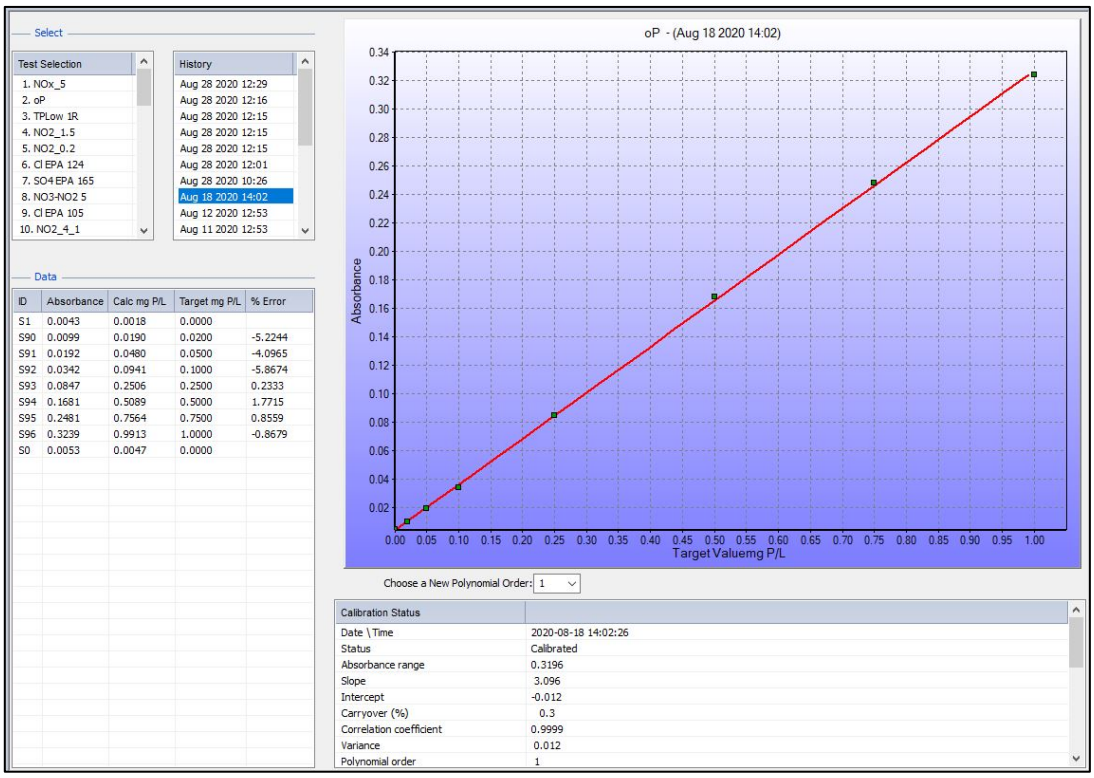
13 STANDARDIZATION



Max. Blank Absorbance (S1): 0.01 AU

Sensitivity Range (1 mg P/L): 0.26 – 0.39 AU

DEFINITIONS:

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover: Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A-B)}{C} \times 100\%$$

Where, A = absorbance of carryover blank (S0)

B = absorbance of blank (S1)

C = calibration absorbance range

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14 PERFORMANCE VALIDATION DATA

14.1 DETECTION LIMIT STUDY

Replicates of 0.005 mg P/L o-phosphate standard in reagent water***

Concentration (mg P/L)

	0.0063	0.0056	0.0049	0.0051
	0.0050	0.0044	0.0041	0.0049
	0.0042	0.0051	0.0043	0.0046
	0.0039	0.0047	0.0041	0.0044
	0.0045	0.0040	0.0046	0.0044
	0.0041	0.0040	0.0040	0.0054
	0.0044	0.0039	0.0044	0.0054
	0.0049	0.0040	0.0041	0.0051
	0.0049	0.0042	0.0043	0.0046
	0.0052	0.0049	0.0042	0.0043
	0.0048	0.0051	0.0043	0.0047
	0.0052	0.0055	0.0052	0.0051
	0.0048	0.0039	0.0051	0.0047
	0.0056	0.0040	0.0045	0.0052
	0.0062	0.0045	0.0055	0.0058
Statistics				
Mean	0.0049	0.0045	0.0045	0.0049
Std. Deviation	0.0007	0.0006	0.0005	0.0004
MDL*	0.0018	0.0016	0.0012	0.0011

14.2 PERFORMANCE STUDY

Replicates of 0.05 mg P/L o-phosphate standard in reagent water***

Concentration (mg P/L)

	0.0468	0.0448	0.0512	0.0519
	0.0461	0.0461	0.0511	0.0500
	0.0493	0.0451	0.0528	0.0514
	0.0469	0.0452	0.0542	0.0504
	0.0479	0.0465	0.0534	0.0501
	0.0487	0.0474	0.0526	0.0501
	0.0488	0.0448	0.0525	0.0519
	0.0484	0.0461	0.0538	0.0492
	0.0492	0.0472	0.0538	0.0496
	0.0492	0.0479	0.0557	0.0515
Statistics				
Mean	0.0481	0.0461	0.0531	0.0506
Std. Deviation	0.0012	0.0011	0.0014	0.0001
% RSD**	2.4	2.4	2.6	1.9
% Recovery	96.2	92.3	106.2	101.2

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14.3 PRECISION STUDY

Replicates of 0.5 mg P/L o-phosphate standard in reagent water***

Concentration (mg P/L)

Statistics				
Mean	0.503	0.525	0.501	0.524
% Recovery	100.7	104.9	100.1	104.7
Std. Deviation	0.008	0.013	0.008	0.007
% RSD**	1.5	2.6	1.6	1.3
Maximum	0.521	0.544	0.516	0.532
Minimum	0.492	0.498	0.489	0.510

Replicates of 1.0 mg P/L o-phosphate standard in reagent water***

Concentration (mg P/L)

Statistics				
Mean	0.985	1.033	0.986	1.053
% Recovery	98.5	103.3	98.6	105.3
Std. Deviation	0.009	0.019	0.010	0.017
% RSD**	1.0	1.9	1.0	1.6
Maximum	1.005	1.058	0.998	1.072
Minimum	0.974	0.996	0.968	1.012

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9.

15 REVISIONS

November 2014	Rev. 0	Method created and support data acquired
April 2020	Rev. 1	Minor editorial changes
		Added sensitivity and blank absorbance information
September 2020	Rev. 2	Update calibration curve and performance study

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